



**DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
NORTH SOUTH UNIVERSITY**

Senior Design Project

Shotota, a Bangla plagiarism tool to Ensure Originality and Promote Honesty.

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LETTER OF TRANSMITTAL

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To

Dr. Rajesh Palit

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: **Submission of Capstone Project Report on "Shotota, a Bangla plagiarism tool to Ensure Originality and Promote Honesty."**

Dear Sir,

With due respect, we would like to submit our Capstone Project Report on "**Shotota, a Bangla plagiarism tool to Ensure Originality and Promote Honesty**" as a part of our BSc program. The report deals with Bangla plagiarism detection System. This project was very much valuable to us as it helped us gain experience from NLP field and apply in real life. We tried to the maximum competence to meet all the dimensions required from this report.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

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DECLARATION

Project Title Shotota, a Bangla plagiarism tool to Ensure Originality and Promote Honesty.
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This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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ABSTRACT

This project emerged in response to the absence of a robust Bangla plagiarism detection tool, a gap that became evident given the richness of the Bangla language. Recognizing the need for a tool comparable to established platforms like Turnitin and Grammarly, we leveraged the state-of-the-art Bangla-BERT model to generate vector embeddings for all sentences in the corpus. The data from the corpus were efficiently pre-processed so that the model could efficiently parse it. These embeddings were then stored in a Qdrant vector database, known for its robust indexing algorithm that facilitates rapid similarity searches. By employing the Bangla-BERT model, we successfully identified and categorized instances of paraphrased plagiarism. Our evaluation metrics revealed that highly paraphrased sentences exhibited a similarity range between 60% to 80%, moderately paraphrased sentences scored between 80% to 90%, and simpler forms of paraphrasing, such as structural paraphrasing, demonstrated similarity scores exceeding 90%. This approach not only addresses the deficiency in Bangla plagiarism detection tools but also introduces an effective solution for discerning nuanced levels of paraphrased content within the intricacies of the Bangla language. Our system boasts a user-friendly interface and facilitates the downloading of comprehensive reports for each document queried. Our overarching goal is to optimize the entire system to align with industry standards, ensuring efficiency and reliability at par with industry-level expectations.

Keywords: Bangla Language, Copy, Indexing, Plagiarism Analysis, Plagiarism, Vector Embedding, Paraphrase.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

Bengali, with approximately 273 million speakers, is the seventh most spoken language in the world. It is a language of immense richness, dating back to prehistoric times. The absence of a reliable Bengali plagiarism detection tool is sad for the Bengali community, particularly in Bangladesh.

Our academic and professional landscape needs a dedicated Bangla plagiarism detection tool. The lack of such a tool has adversely affected the educational landscape in Bangladesh, as journals struggle to identify plagiarism in academic papers, and students often get away with copying content from external sources without consequences. "As reported by tbsnews.net, plagiarism incidents in Bangladesh's universities are on the rise due to the unavailability of a reliable plagiarism detection tool." Book publishers also face challenges as they have to check for plagiarism manually, resulting in a time-consuming and laborious process. This situation poses a significant obstacle to maintaining academic integrity and fairness within the Bengali community. "Copyright violations and plagiarism are affecting the overall quality of higher education in Bangladesh, according to aa.com.tr."

To address this issue, our plagiarism detection tool aims to provide a comprehensive platform that caters to the specific needs of various domains. Because Current tools focus on significant languages, leaving a critical gap for those working extensively in Bangla.

1.2 Purpose and Goal of the Project

Through our plagiarism detection tool, educational institutions can effectively detect plagiarism, while book publishers can automate their plagiarism-checking processes. Additionally, journals can ensure the authenticity of content before publication. We aim to offer a tailored system for each domain,

addressing their unique requirements and enhancing the integrity of academic and literary works within the Bengali community.

1.3 Organization of the Report

In Chapter 2 of the report, we have shown existing research and limitations related to this project. In Chapter 3, we discussed the methodology, which contains system design, hardware and software components needed for the project, and how we have implemented the hardware and software system. In Chapter 4, we have shown our experiment results on the project and demonstrated an overall analysis of the project. In Chapter 5, we have discussed the project's impacts on societal, health, legal, and cultural issues. Moreover, it's impact on environment and sustainability. In Chapter 6, we discussed the project planning and budget needed to complete the project on time. In Chapter 7, we discussed all the complex engineering problems and activities related to this project. In Chapter 8, we have concluded our report with a summary, limitations, and future project improvements.

Chapter 2

Research Literature Review

2.1 Existing Research

In this article [1], the authors discussed the fact that the art in evaluating plagiarism detection which has not been studied systematically until now. They analyzed experiment tasks, performance measures, and underlying corpora. They found that the majority of the approaches perform overlap detection by Exhaustive comparison against some locally stored document collection. They also discussed the need for an open, commonly used plagiarism detection corpus. They gave an overview of the PAN-PC-10 corpus for the controlled evaluation of detection algorithms. Moreover, their evaluation framework has been used by 31 plagiarism detectors for comparison purposes.

In this article [2], the authors discussed the BERT NLP model which is the first-ever deeply bidirectional and fully unsupervised technique for language representation that was pre-trained using a plain text corpus. The article addresses Bangla-Bert's deficiencies and a proposed monolingual BERT for Bangla language based on a large dataset named BanglaLM. The dataset is 40GB, with three variants containing around 20 million samples for each variant. It's an enormous corpus of internet sources, including news, blog sites, government journals etc. After training on dataset BanglaLM Bangla-BERT precision and recall respectively reached 0.9989 and 0.9999 for NER (Named Entity Recognition) on the Bengali NER corpus, on CROSS LINGUAL SENTIMENT ANALYSIS Bangla-BERT reached 0.9703 accuracy and A 0.93 F1 score, also Bangla-BERT's 0.979 AUC score is the best overall performance among all other methods.

In this article [3], the authors discussed the importance of text mining and preprocessing techniques for NLP based research purposes. The article addresses different pre-processing techniques such as: truncating methods, statistical methods, mixed methods, corpus-based stemmer, Term

Frequency-Inverse Document Frequency etc. It also discussed the disadvantages of a few techniques.

In this article [4], the authors extensively examined the impact of preprocessing on text classification in terms of various aspects such as classification accuracy, text domain, text language, and dimension reduction. They came to the conclusion that choosing appropriate combinations of preprocessing tasks, rather than enabling or disabling them all, may provide significant improvement on classification accuracy depending on the domain and language studied.

In this article [5], the authors discussed having a freely available architecture for plagiarism detection across languages covering the entire process: heuristic retrieval, detailed analysis, and post-processing. They used three cross-language similarity estimation models under the same conditions on the different plagiarism detection sub-tasks. The authors came to the conclusion that among the three models (CL-ASA, CL-CNG, T+MA) T+MA produces the best results closely followed by CL-ASA showing higher values of precision.

The paper [6] presents a proposed method for developing a Bangla plagiarism detection system. The corpus consists of all books of the National Curriculum and Textbook Board (NCTB) of Bangladesh from classes I-XII. The corpus is processed through text extraction, tokenization, stop words removal, and TF-IDF scores generation. The authors use the TF-IDF algorithm to vectorize both the suspicious text and the corpus texts, then apply the Cosine Similarity algorithm to calculate the similarity scores. If the similarity score exceeds a specific threshold, the system identifies the text as plagiarized and provides a plagiarism report indicating the source document. The authors claim a high accuracy rate of 96.75

The paper [7] addresses the problem of plagiarism by discussing methods for its reduction and prevention, specifically focusing on detection techniques. The paper also highlights the different plagiarism methods, such as copy-paste, paraphrasing, translation, artistic plagiarism, idea plagiarism, code plagiarism, improper use of quotation marks, and misinformation of references. Plagiarism reduction methods can be classified into two categories: prevention and detection. Prevention methods, promoting a shift in societal attitudes towards plagiarism, are considered the most effective means of addressing the problem. Detection methods involve using software tools to identify instances of plagiarism automatically. Regarding plagiarism detection, the authors provide insights into various similarity metrics, such as cosine similarity and Euclidean distance.

The paper [8] shows an approach to plagiarism detection using sentence-based document comparison. The algorithm first pre-processes the data. It converts data to lowercase, removes common words that add no meaning, and eliminates repeated words within a sentence. the algorithm compares the processed documents by computing a similarity score based on the word count metric. The similarity score is the average similarity between sentences, considering common words and lengths.

The paper [9] provides a comprehensive overview of plagiarism detection systems, explores various forms of plagiarism, and discusses the methods and tools used to detect plagiarism. It considers plagiarism are copy-paste plagiarism, paraphrasing plagiarism, metaphor plagiarism, idea plagiarism, recycled plagiarism, 404 Error/Illegitimate Source plagiarism, and re-tweet plagiarism. Source code plagiarism involves manipulation, reordering structure, no change, and language switching. The paper provides a list of plagiarism detection tools, and its year of development, characteristics, and functionality. It highlights the advancements in plagiarism detection tools, the importance of datasets, and the challenges involved in plagiarism detection.

The paper [10] discusses the classification of plagiarism types and the different types of plagiarism detection systems. Not all types of plagiarism are equally difficult for computerized plagiarism detectors. For example, verbatim copying can be easily detected using string matching, while paraphrasing requires natural language processing methods to identify similar assertions in the source and plagiarized texts. Regarding plagiarism detection systems, they are categorized as hermetic or web-based, and general purpose, natural language, or source code oriented. Hermetic systems search for plagiarism within a local collection of documents, while web-based systems search for matches in online sources. Some systems combine both web and hermetic detection. The current state of the art in plagiarism detection systems has advanced from simple verbatim copying detection to more sophisticated methods. Modern systems can now detect advanced types of plagiarism, including approximate string matching to find similarities between documents, even if there are changes in sentence structure or word substitution. Advancements have been made in plagiarism detection systems, but challenges remain in detecting complex forms of plagiarism and addressing ethical and legal concerns.

This paper [11] presents the development of a learning model called Team Re-CIP to address the issue of academic plagiarism among students in universities. It followed the ADDIE development

model. ADDIE stands for Analysis, Design, Development, Implementation, and Evaluation. The learning model was based on a needs assessment conducted with 208 students, which revealed a high incidence of plagiarism among students. The model underwent validation by learning experts and was refined based on their feedback. The results of product trials indicated that the learning model was reliable, understandable, and applicable.

This paper [12] examines the originality of essays generated by ChatGPT and their potential for plagiarism. The researchers collected 50 essays on various topics generated by ChatGPT and evaluated their originality using plagiarism detection tools. The results show that ChatGPT can generate original content that is difficult to detect as plagiarized. ChatGPT showed better performance in identifying whether the essays were generated by it. The paper tells us to reconsider our traditional plagiarism detection system.

In this article [13], the authors discussed how AI generated chatboxes may be utilized to fabricate research in the medical community, with examples. Furthermore, they compared studies of human detection of AI-based works to gauge the accuracy of identification of fabricated, AI-generated works. Additionally, they tested the accuracy of free, online AI detectors. The danger of fabricated research is then highlighted, along with the reasons why one would want to fabricate medical research and potential remedies to this looming threat. They foresee that these issues will present themselves as AI technologies continue to expand in quantity and quality, and they discussed to begin an initial discussion on how to better develop and implement safeguards against this threat to the medical community.

In this article [14], the authors proposed two methods to identify Extrinsic plagiarism. In both methods, to limit the search space, two stages of filtering based on the bag of word (BoW) technique are used at the document level and at the sentence level, and plagiarism is investigated only in the outputs of these two stages. In the first method to detect similarities in suspicious documents and sentences, they used combination of pre-trained network technique of words embedding Fast-Text and TF-IDF weighting technique to form two structural and semantic matrices and in the second method to form the two matrices, they used WordNet ontology and weighting TF-IDF. They used dice similarity and the structural similarity of the weighted composition, to calculate two similarity values. Then they compared the similarity of suspicious sentences with the minimum threshold, the document containing the suspicious sentence receives the label of plagiarism or non-plagiarism. Their experimental results on the PAN-PC-11 database show that the first method has achieved 95.1%

precision and the second method 93.8% precision, which shows that the use of word embedding network compared to WordNet ontology can be more successful in detecting Extrinsic plagiarism.

In this article [15], the authors introduces APlag, a tool to find plagiarism in Arabic texts. Plagiarism happens when someone uses someone else's work without giving credit. This tool helps catch that. They made APlag by looking at how Arabic works and using smart ways to compare texts. They changed the words to make it easier to see if someone copied. They tested it with different texts and found it's very good at finding plagiarism. They also compared it to another tool, APD, and showed that APlag finds more copied sentences. APlag is useful because it can catch when people change words to hide copying. They want to do more tests on bigger texts to make APlag even better. Overall, APlag is a good tool for finding plagiarism in Arabic texts by finding even tricky changes in words.

2.2 Limitations

Despite the growing importance of addressing plagiarism in the Bengali community, existing research on plagiarism detection tools for Bengali exhibits several limitations.

2.2.1 Language Specificity

A big problem is that there aren't many tools that can find copied stuff in Bengali writing. Most tools focus on English and don't work well with Bengali because the language is different, with its own special ways of saying things. This creates a gap, making it tough to find copied content accurately in Bengali writing.

2.2.2 Limited Coverage of Plagiarism Types

The current studies mostly look at basic types of copying, like copy-pasting and rephrasing. They don't pay much attention to more complex types of copying, like stealing ideas, copying art, or messing with references. This makes it tough to find a good solution to catch all kinds of copying in Bengali content.

2.2.3 Scalability and Efficiency

Some studies haven't looked enough at how well plagiarism tools work when dealing with lots of Bengali text. It's important to see how these tools perform in real-time, especially in universities or

publishers where many documents are checked quickly. This needs more investigation to make sure the tools are fast and efficient.

2.2.4 Challenges in Cross-Language Plagiarism Detection

Many plagiarism detection tools are made to find copied content in just one language. They may struggle to catch plagiarism when people mix different languages or translate text. This is a problem, especially when dealing with Bengali content that might include words from various languages. As a result, it becomes harder to accurately spot plagiarism that comes from different language backgrounds.

2.2.5 Limited Evaluation Metrics

The evaluation metrics employed in existing studies may not fully capture the effectiveness of plagiarism detection tools in real-world scenarios. The lack of diverse and comprehensive evaluation criteria hampers the understanding of a tool's practical utility.

2.2.6 Lack of Diverse Data Sources

The current studies usually use only a small set of sources in Bengali, not considering a variety of different ones. This makes it hard for plagiarism checkers to work well in different situations.

Chapter 3

Methodology

3.1 System Design

Using vector embeddings and keyword-based web scraping, we concentrated on building a system that could efficiently identify paraphrased phrases and check for plagiarism. The main need was to develop a plagiarism detection system that could properly recognize paraphrased lines. Vector embeddings, which can identify semantic similarities between texts, have to be used as a result of this. Additionally, the system searches the internet for extra information not found in the system corpus.

There isn't a trustworthy plagiarism detector available in Bangla that can be utilized for both academic and non-academic applications. Since this system seeks to be the first plagiarism system that is capable of detecting plagiarism in Bangla at a high level, there is a need for a reliable, quick, accurate, and user-friendly system.

We thoroughly examined the shortcomings of the current plagiarism detection tools. Many systems, frequently missing subtle paraphrasing, only used string matching or simple keyword-based techniques. For the Bangla language, there isn't a reliable plagiarism detector like Turnitin. As was already indicated, current algorithms that check for plagiarism rely on string matching methods that cannot recognize paraphrased phrases. By using vector embeddings to detect semantic similarities, our method intended to get beyond these restrictions and offer a more complete plagiarism detection solution.

Since there are more and more texts being written in Bangla every day, our system's corpus will likewise grow everyday. The system was created with scalability in mind to handle massive amounts of data as a result. The system is capable of handling massive volumes of data with accurate and

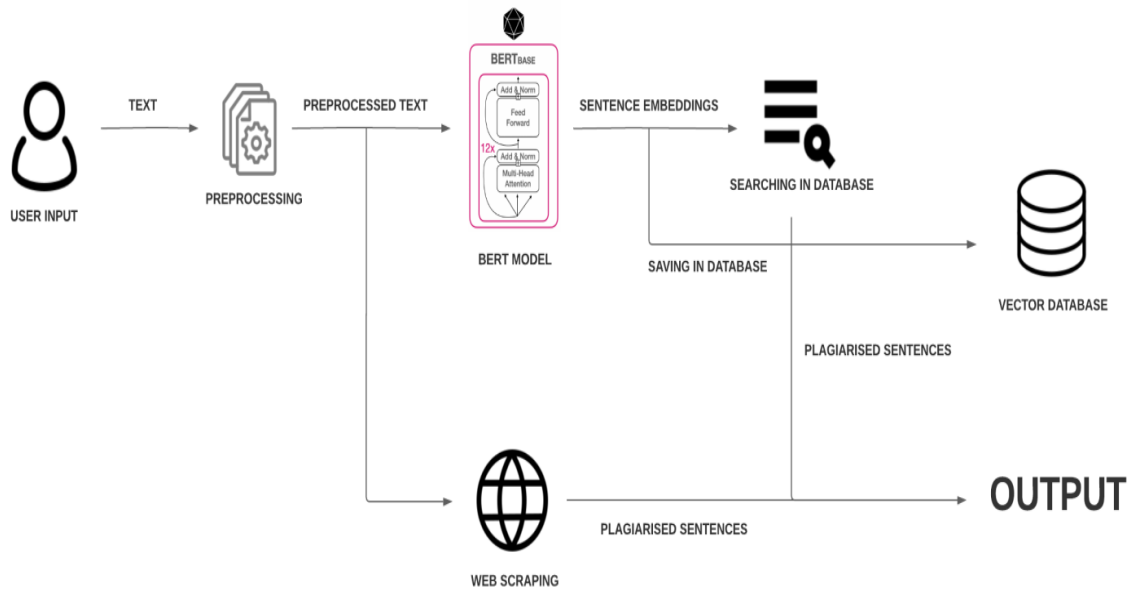


Figure 3.1: System

effective indexing.

During site scraping and text addition to our own corpus, we complied with pertinent norms and laws pertaining to data privacy, intellectual property rights, and ethical issues. Regarding the acquisition and processing of online and offline content, we made sure that our system conformed with all applicable legal and ethical requirements.

We had to choose between accuracy, speed, and computational resources during the design process. By utilizing sophisticated vector embeddings, we put accuracy first while also tuning the system to provide results quickly enough. With our present resources, we thoughtfully weighed these trade-offs to achieve a balance between efficacy and efficiency.

All design choices, such as the choice of certain vector embedding models, the runtime method for web scraping, and the indexing strategies used, were fully described. We wanted to promote future knowledge, replication, and prospective system enhancements through the documentation of our design decisions. By taking into account these design factors, we made sure that a reliable method for detecting plagiarism was created that could efficiently capture paraphrased lines.

3.2 Software Components

3.2.1 Django Framework

Django is a strong web tool that uses Python. It helps make big and easy-to-manage websites. It follows the Model-View-Controller (MVC) architectural pattern. Django simplifies the creation of complex, database-driven websites by providing a high-level, Pythonic interface.

3.2.2 BERT Model

We employ the BERT (Bidirectional Encoder Representations from Transformers) model to better understand and extract features from natural language. BERT is a competent pre-trained language model that grasps the connections and meanings of words in a sentence. In this project, we use Bangla BERT by Sagor Sarker, designed for the Bangla language, ensuring accurate processing and performance improvements in Bangla language vector representations.

3.2.3 Qdrant

To enable efficient similarity searching and retrieval of textual data, a vector database will be employed. The proposed database is Qdrant. It is a free-to-use tool that helps find similar things in big sets of information. It is built to manage large amounts of data effectively, making it suitable for applications with a lot of information. It's convenient when dealing with data represented as vectors, which are numerical ways of showing items. In simpler terms, Qdrant helps quickly find things alike in large data sets.

Tool	Functions	Other similar Tools (if any)	Why selected this tool
Django Framework	Simplifies web development	Laravel	Robust Pythonic interface
BERT Model	Enhances NLP tasks		Captures contextual relationships
Qdrant	Efficient similarity searching, handles large datasets	Milvus	Works well with numerical representations

Table 3.1: List of Software Tools.

3.3 Software Implementation

3.3.1 Preprocessing

In the preprocessing step of the project, the textual data undergoes several transformations to improve its quality and relevance for further analysis. The process includes removing punctuation marks and numerical digits, eliminating stop-words, removing unnecessary spaces between words, and filtering out non-Bangla sentences. These preprocessing techniques ensure that the data is cleaned and prepared appropriately for subsequent tasks, such as feature extraction, modeling, and analysis.

3.3.2 BERT Model

To achieve effective natural language understanding and feature extraction, the proposed method involves utilizing the BERT (Bidirectional Encoder Representations from Transformers) model. BERT is a powerful pre-trained language representation model that can capture the contextual relationships and semantics of words in a sentence. By fine-tuning BERT on specific tasks, such as text classification or sentiment analysis, it can be used to enhance the accuracy and performance of various natural language processing tasks in this project.

3.3.3 Vector Database

To enable efficient similarity searching and retrieval of textual data, a vector database will be employed. The proposed method includes using Milvus, a high-performance vector database specifically designed for similarity search tasks. Milvus provides fast indexing and querying capabilities for high-dimensional vectors, which will be beneficial for storing and retrieving feature representations generated by the BERT model in this project.

3.3.4 Plagiarism Detection Algorithm

To detect instances of plagiarism, an algorithm tailored for plagiarism detection will be developed. The algorithm will utilize the preprocessed textual data, feature representations from the BERT model, and appropriate similarity measures to compare and identify potential instances of plagiarism. The algorithm will consider various factors, such as the similarity threshold, document size, and specific plagiarism detection techniques, to achieve accurate and reliable plagiarism detection results.

3.3.5 User Interface

To provide an interactive and user-friendly experience, a user interface will be developed as part of the proposed methods. The user interface will allow users to interact with the plagiarism detection system, input text or files to be analyzed, and receive plagiarism reports or notifications. The user interface will be designed to be intuitive, visually appealing, and provide informative feedback to users regarding the detected plagiarism and related details.

3.3.6 Alternate Methods

In an alternate approach, the proposed method suggests using Qdrant as the vector database for efficient similarity searching and retrieval of textual data. Qdrant is an open-source vector search engine that provides a robust and flexible solution for storing and querying high-dimensional vectors. It offers various distance metrics, powerful indexing mechanisms, and advanced querying capabilities. By incorporating Qdrant, the project can benefit from its speed, scalability, and rich feature set to enhance the efficiency and effectiveness of the plagiarism detection system.

3.4 Evaluation Criteria

3.4.1 Accuracy

Measure the accuracy of our tool by comparing the detected instances of plagiarism against manually annotated or known instances of plagiarism in a labeled dataset. Calculate metrics such as precision, recall, and F1 score to assess the tool's ability to correctly identify plagiarism.

3.4.2 Sensitivity

Evaluate the sensitivity of our tool by analyzing its ability to detect different types of plagiarism, such as verbatim copying, paraphrasing, or translation-based plagiarism. Use a diverse dataset with various plagiarism scenarios to determine the tool's sensitivity to different forms of plagiarism.

3.4.3 Speed and Efficiency

Measure the processing speed and efficiency of our tool by recording the time taken to process different document sizes. Assess how quickly the tool can analyze and identify instances of plagiarism, considering real-time or near-real-time detection requirements.

3.4.4 Scalability

Evaluate the scalability of your tool by testing its performance with increasing dataset sizes. Assess how the tool's response time and accuracy are affected as the dataset grows, ensuring that it can handle large-scale plagiarism detection tasks.

3.4.5 Coverage or Source Diversity

Assess the coverage or source diversity of our tool by using a diverse corpus of Bangla documents from different domains or genres. Ensure that your tool can effectively detect plagiarism across various sources, including academic papers, news articles, and online content.

3.4.6 Privacy and Security

Ensure that your tool adheres to privacy and security standards. Assess how the tool handles user data and sensitive information, ensuring that plagiarism detection processes are conducted securely and confidentially.

3.4.7 Flexibility and Customization

Evaluate the flexibility and customization options of your tool. Assess whether it allows users to adjust parameters, configure settings, or adapt the tool to specific requirements, ensuring its adaptability to different use cases.

3.4.8 Reporting and Documentation

Assess the reporting and documentation capabilities of your tool. Evaluate whether it provides clear and comprehensive reports of detected plagiarism, including detailed information about the matched sources, similarity scores, and highlighted plagiarized sections.

3.4.9 Continuous Improvement and Updates

Consider the tool's potential for continuous improvement and updates. Assess whether it offers mechanisms for incorporating new algorithms, enhancing accuracy, and addressing emerging plagiarism patterns or techniques.

3.4.10 User-Friendliness

Evaluate the user-friendliness of our tool by conducting usability tests or gathering feedback from users. Consider factors such as ease of installation, intuitive user interface, clear instructions, and overall user satisfaction.

Chapter 4

Result and Discussion

Due to limited resources on Google Colab, we had to chunk the 27 GB corpus into 40 MB files. We then used the Bangla-BERT model to generate vector embeddings for each of these files. The process of generating vector embeddings for a single 40 MB file took approximately one hour. Each 40 MB file contained around 200,000 sentences. The name of the corpus we are using is Bangla2B+. The table below shows the estimated time required to fully vectorize the whole corpus.

During the preprocessing of the corpus, we tokenize each sentence and apply several steps. These

Corpus Size	Number of Sentences	Estimated Time
27 GB	138 million	691 hours

Table 4.1: Index and Search time based on nlist and nprobe.

steps involve removing punctuation marks and numerical digits, eliminating stop-words, removing unnecessary spaces between words, and filtering out non-Bangla sentences. It takes approximately 0.0199 seconds to preprocess every 1000 Bangla sentence.

For our project, we utilized two vector databases: Milvus and Qdrant. In Milvus, the insertion time for 1000 vectors was 3.33 seconds for HNSW indexing and 3.32 seconds for IVF_FLAT indexing. In Qdrant, the insertion time for 1000 vectors was also 8.02 seconds. In Milvus, the indexing time for the HNSW algorithm was 0.602 seconds, while for the IVF_FLAT algorithm, it took 0.619 seconds. On the other hand, in Qdrant, there is no separate time needed for indexing as the indexing process is automatically done when the vector database is created.

In Table 2, we present the search times observed for various combinations of nprobe and nlist values in the HNSW algorithm implemented in Milvus. The search times are measured in seconds.

In Table 3, we did the same thing for the IVF_FLAT algorithm implemented in Milvus.

Further evaluation of the system will be conducted once it is fully ready with a 100 GB corpus of embeddings stored in a database. The system will be fully evaluated on every criterion before being

Idx \Src	1	2	4	8	16	32	64	128	256	512
1	0.6 \0.02	0.61 \0.018	0.57 \0.02	0.61 \0.02	0.61 \0.019	0.61 \0.019	0.61 \0.018	0.61 \0.02	0.61 \0.02	0.57 \0.019
2	0.61 \0.02	0.61 \0.017	0.61 \0.02	0.61 \0.019	0.61 \0.019	0.61 \0.02	0.57 \0.02	0.61 \0.02	0.61 \0.019	0.61 \0.02
4	0.57 \0.02	0.61 \0.019	0.57 \0.02	0.61 \0.019	0.57 \0.019	0.61 \0.019	0.61 \0.019	0.57 \0.018	0.61 \0.02	0.57 \0.02
8	0.61 \0.02	0.61 \0.02	0.61 \0.018	0.61 \0.019	0.57 \0.018	0.61 \0.019	0.57 \0.02	0.61 \0.019	0.61 \0.02	0.57 \0.02
16	0.61 \0.019	0.67 \0.02	0.61 \0.02	0.61 \0.02	0.62 \0.02	0.61 \0.019	0.61 \0.02	0.57 \0.018	0.61 \0.02	0.57 \0.02
32	0.61 \0.02	0.61 \0.02	0.61 \0.02	0.61 \0.018	0.64 \0.02	0.6 \0.019	0.61 \0.02	0.6 \0.02	0.57 \0.02	0.61 \0.021
64	0.55 \0.019	0.61 \0.019	0.52 \0.013	0.52 \0.013	0.52 \0.012	0.52 \0.012	0.53 \0.013	0.52 \0.012	0.52 \0.013	0.52 \0.012
128	0.52 \0.012	0.52 \0.012	0.52 \0.012	0.52 \0.012	0.52 \0.012	0.56 \0.013	0.57 \0.013	0.52 \0.014	0.52 \0.012	0.52 \0.012
256	0.52 \0.012	0.56 \0.013	0.52 \0.012	0.52 \0.012	0.52 \0.012	0.56 \0.013	0.52 \0.012	0.52 \0.013	0.56 \0.012	0.52 \0.012
512	0.52 \0.012	0.52 \0.014	0.52 \0.013	0.52 \0.013	0.52 \0.013	0.53 \0.013	0.52 \0.012	0.56 \0.014	0.52 \0.013	0.52 \0.013

Table 4.2: Index and Search time based on nlist and nprobe.

launched. Through this rigorous evaluation, any necessary improvements and enhancements can be identified, further refining the system’s capabilities and overall performance. The best results are

Idx \Src	1	2	4	8	16	32	64	128	256	512
1	0.59 \0.017	0.59 \0.021	0.56 \0.019	0.59 \0.02	0.56 \0.019	0.59 \0.02	0.57 \0.02	0.6 \0.02	0.56 \0.02	0.61 \0.02
2	0.56 \0.02	0.56 \0.02	0.57 \0.019	0.61 \0.019	0.62 \0.019	0.59 \0.02	0.59 \0.019	0.56 \0.02	0.59 \0.02	0.6 \0.019
4	0.59 \0.019	0.59 \0.019	0.59 \0.019	0.59 \0.02	0.56 \0.02	0.56 \0.019	0.59 \0.018	0.59 \0.02	0.59 \0.019	0.5* \0.019
8	0.59 \0.019	0.59 \0.019	0.6 \0.02	0.6 \0.02	0.56 \0.019	0.59 \0.018	0.56 \0.02	0.59 \0.019	0.59 \0.019	0.56 \0.019
16	0.59 \0.012	0.59 \0.019	0.60 \0.02	0.56 \0.018	0.56 \0.02	0.59 \0.019	0.59 \0.02	0.59 \0.019	0.6 \0.02	0.6 \0.02
32	0.57 \0.02	0.56 \0.02	0.6 \0.019	0.56 \0.02	0.61 \0.019	0.56 \0.019	0.56 \0.02	0.59 \0.02	0.6 \0.018	0.6 \0.018
64	0.6 \0.019	0.61 \0.019	0.6 \0.018	0.56 \0.013	0.52 \0.012	0.53 \0.013	0.54 \0.012	0.52 \0.013	0.52 \0.013	0.53 \0.012
128	0.53 \0.012	0.53 \0.013	0.52 \0.013	0.54 \0.013	0.54 \0.013	0.53 \0.013	0.53 \0.012	0.52 \0.013	0.57 \0.013	0.53 \0.013
256	0.54 \0.013	0.53 \0.013	0.52 \0.012	0.53 \0.012	0.55 \0.013	0.56 \0.012	0.53 \0.012	0.53 \0.013	0.54 \0.013	0.53 \0.012
512	0.53 \0.012	0.53 \0.013	0.52 \0.011	0.54 \0.013	0.56 \0.013	0.56 \0.012	0.53 \0.012	0.54 \0.013	0.53 \0.013	0.57 \0.012

Table 4.3: Index and Search time based on nlist and nprobe.

marked in red.

As we know there are many forms of paraphrasing one of which is splitting one sentence into multiple short sentence. So, here we have split the original sentence into multiple short paraphrased sentences shown in figure 4.1 and 4.2 to simulate these type of situation. Here we applied four type of preprocessing on the paraphrased sentences. Here in P1 preprossessing we removed punctuation’s from sentences, in P2 preprocessing we removed punctuation’s and also stemmed the words into smaller forms, in P3 preprocessing we kept each sentences in the original form containing punctuation’s, lastly in P4 preprocessing we preprocessed in our own manner which resulted in sentences without punctuation’s and words from other language. According to our figure, the original sentences in this case are texts 1 through 7, and the split paraphrased short sentences are S1 through S4. The similarity scores for text 1 range from **0.77** to **0.7978**. The similarity scores in text 2 then range from **0.81** to **0.8521**. The similarity scores in text 3 then range from **0.82** to **0.83**. Similarity scores in text 4

Text1: আমি একটি প্রিয় বই পড়ছি যা বড়ই মজাদার স্বস্তিকর এবং শিক্ষামূলক।
S1 = আমি একটি প্রিয় বই পড়ছি
S2 = আমি একটি প্রিয় বই পড়ছি
S3 = আমি একটি প্রিয় বই পড়ছি।
S4 = আমি একটি প্রিয় বই পড়ছি

Text2: আমি একটি প্রিয় বই পড়ছি যা বড়ই মজাদার স্বস্তিকর এবং শিক্ষামূলক
S1 = এটি বড়ই মজাদার স্বস্তিকর এবং শিক্ষামূলক
S2 = এটি বড়ই মজাদার স্বস্তি এবং শিক্ষা
S3 =এটি বড়ই মজাদার, স্বস্তিকর, এবং শিক্ষামূলক।
S4 =এটি বড়ই মজাদার স্বস্তিকর এবং শিক্ষামূলক3))

Text3: আমি বই পড়ছি এবং আমার বন্ধু গান শুনছে
S1 = আমি বই পড়ছি
S2 = আমি বই পড়ছি
S3 = আমি বই পড়ছি।
S4 =আমি বই পড়ছি

Text4: 'আমি বই পড়ছি এবং আমার বন্ধু গান শুনছে
S1 = আমার বন্ধু গান শুনছে
S2 = আমি বন্ধু গা শুন
S3 =আমার বন্ধু গান শুনছে।
S4 = আমার বন্ধু গান শুনছে

Figure 4.1: Original sentences split into multiple sentences (1)

again range from **0.76** to **0.85**. Once more, scores in text 5 range from **0.56** to **0.62**. Scores in texts 6 and 7 range from **0.7** to **0.72** and **0.73** to **0.80**, in that order. Since the similarity scores fall inside our evaluation parameters, as can be shown, our algorithm is able to identify plagiarism in this sort of circumstance.

	S1	S2	S3	S4
Text1	0.7850	0.7978	0.7787	0.7850
Text2	0.8496	0.8154	0.8521	0.8496
Text3	0.8277	0.8204	0.8300	0.8277
Text4	0.8286	0.7686	0.8541	0.8286
Text5	0.5659	0.6266	0.6130	0.5659
Text6	0.7209	0.7052	0.7212	0.7209
Text7	0.7714	0.7389	0.8068	0.7714

Table 4.4: Result

Here for every original sentences we have taken four different paraphrased sentences such as S1, S2, S3, S4 and for every paraphrased sentences we applied four different types of prepossessing such as P1,P2,P3,P4. Here in P1 prepossessing we removed punctuation's from sentences, in P2 preprocessing we removed punctuation's and also stemmed the words into smaller forms, in P3 preprocessing we kept each sentences in the original form containing punctuation's, lastly in P4

Text5: তিনি চলছেন শপিং করতে প্রিয়জনের সাথে সময় কাটাচ্ছেন এবং রেস্টোরাঁয় ভোজন করছেন

S1 = তিনি চলছেন শপিং করতে

S2 = সে চল শপিং করত

S3 = তিনি চলছেন শপিং করতে।

S4 =তিনি চলছেন শপিং করতে

Text 6: তিনি চলছেন শপিং করতে প্রিয়জনের সাথে সময় কাটাচ্ছেন এবং রেস্টোরাঁয় ভোজন করছেন

S1 = প্রিয়জনের সাথে সময় কাটাচ্ছেন

S2 = প্রিয়জ সাথ সময় কাট

S3 =প্রিয়জনের সাথে সময় কাটাচ্ছেন।

S4 = প্রিয়জনের সাথে সময় কাটাচ্ছেন

Text7: তিনি চলছেন শপিং করতে প্রিয়জনের সাথে সময় কাটাচ্ছেন এবং রেস্টোরাঁয় ভোজন করছেন

S1 = এবং রেস্টোরাঁয় ভোজন করছেন

S2 = এবং রেস্টোরাঁয় ভোজন কর

S3 =এবং রেস্টোরাঁয় ভোজন করছেন।

S4 =এবং রেস্টোরাঁয় ভোজন করছেন।

Figure 4.2: Original sentences split into multiple sentences (2)

preprocessing we preprocessed in our own manner which resulted in sentences without punctuation's and words from other language. Between four types of preprocessing we can see P2 has the better scores among all other preprocessing techniques such as **1.0, 0.95, 0.90, 0.93, 0.84**. Though P2 shows better scores but there is a technical problem with P2. The library which was used in P2 preprocessing is not well developed hence the stemmed words produced in P2 preprocessing maybe wrong which in result can cause decrease of similarity scores. So, in this aspect P1 is more safer then P2 though it might lose some accuracy but the scores of paraphrased sentence will not decrease as much as for a wrong stemmed word. In most of cases we can see the similarity scores are well above 80 percent which in our system is called a plagiarised sentence. So, any of these preprocessing techniques are good enough to detect plagiarised sentence.

	S1	S2	S3	S4	S5
P1	0.9418	0.9418	1.0000	0.99	0.99
P2	0.9371	0.9371	1.0000	1.00	1
P3	0.9364	0.9364	1.0000	1	0.99
P4	0.9418	0.9418	0.83	0.95	0.80

Table 4.5: Different paraphrasing and preprocessing of sentences (1)

	S1	S2	S3	S4	S5
P1	0.9468	1.0	1.0000	1.00	1.00
P2	0.9548	1.0	1.0000	1.00	1
P3	0.9465	0.96	1.000	1	0.99
P4	0.9468	1.0	0.604	9.03	0.85

Table 4.6: Different paraphrasing and preprocessing of sentences (2)

	S1	S2	S3	S4	S5
P1	0.9034	0.86	0.5701	1.00	1
P2	0.9099	0.87	0.57	1	1
P3	0.8909	0.85	0.85	1	0.99
P4	0.9034	0.86	0.57	0.9033	0.89

Table 4.7: Different paraphrasing and preprocessing of sentences (3)

	S1	S2	S3	S4	S5
P1	0.86	0.86	0.74	0.86	0.89
P2	0.93	0.93	0.86	0.93	0.83
P3	0.88	0.87	0.74	0.88	0.91
P4	0.90	0.86	0.76	0.86	0.88

Table 4.8: Different paraphrasing and preprocessing of sentences (4)

	S1	S2	S3	S4	S5
P1	0.83	0.84	0.76	0.83	0.89
P2	0.84	0.84	0.82	0.84	0.85
P3	0.82	0.82	0.76	0.82	0.88
P4	0.83	0.84	0.74	0.86	0.81

Table 4.9: Different paraphrasing and preprocessing of sentences (5)

Main sentence = ১৮৯৯ খ্রিষ্টাব্দের ২৪ মে (জ্যৈষ্ঠ ১১, ১৩০৬ বঙ্গাব্দ) ভারতের পশ্চিমবঙ্গের বর্ধমান জেলার আসানসোল মহকুমার চুরুলিয়া গ্রামে জন্মগ্রহণ করেন কাজী নজরুল ইসলাম।

S1 = ১৮৯৯ খ্রিষ্টাব্দে, মে মাসের ২৪ তারিখে (বাঙালি পত্রিকাতে তারিখ দেখানো হয়েছে ১১ জ্যৈষ্ঠ, ১৩০৬ বঙ্গাব্দ), ভারতের পশ্চিমবঙ্গের বর্ধমান জেলার আসানসোল মহকুমার চুরুলিয়া গ্রামে কাজী নজরুল ইসলামের জন্ম হয়।

S2 = ১৮৯৯ খ্রিষ্টাব্দে, মে মাসের ২৪ তারিখে (বাঙালি পত্রিকাতে তারিখ দেখানো হয়েছে ১১ জ্যৈষ্ঠ, ১৩০৬ বঙ্গাব্দ), ভারতের পশ্চিমবঙ্গের বর্ধমান জেলার আসানসোল মহকুমার চুরুলিয়া গ্রামে কাজী নজরুল ইসলামের জন্ম হয়।

S3 = কাজী নজরুল ইসলাম ১৮৯৯ সালের ২৪ মে ভারতের পশ্চিমবঙ্গের চুরুলিয়া গ্রামে জন্মগ্রহণ করেন।

S4 = ১৮৯৯ খ্রিষ্টাব্দে, মে মাসের ২৪ তারিখে (বাঙালি পত্রিকাতে তারিখ দেখানো হয়েছে ১১ জ্যৈষ্ঠ, ১৩০৬ বঙ্গাব্দ), ভারতের পশ্চিমবঙ্গের বর্ধমান জেলার আসানসোল মহকুমার চুরুলিয়া গ্রামে কাজী নজরুল ইসলামের জন্ম হয়।

S5 = কাজী নাজরুল ইসলাম জন্মগ্রহণ করেছিলেন ২৩ শে মে ভারতের পশ্চিম বাংলার বুর্দওয়ান জেলার আলদিয়া দে আসানসোল বিভাগে।

Figure 4.3: Different paraphrasing and preprocessing of sentences (1)

Main sentence = তার বাবা ফকির আহমদ ছিলেন স্থানীয় মসজিদের ইমাম এবং মাযারের খাদেম।

S1 = তার বাবা, ফকির আহমদ, স্থানীয় মসজিদের ইমাম এবং মাযারের খাদেম ছিলেন।

S2 = তার বাবা, ফকির আহমদ, ছিলেন স্থানীয় মসজিদের ইমাম এবং মাযারের খাদেম।

S3 = তার বাবা একজন মসজিদ নেতা এবং তার মা পরিবারের দেখাশোনা করতেন।

S4 = তার বাবা, ফকির আহমদ, স্থানীয় মসজিদের ইমাম এবং মাযারের খাদেম ছিলেন।

S5 = তাঁর বাবা ফকির আহমেদ ছিলেন স্থানীয় মসজিদের চৌধুর এবং অভয়ারণের চাকর।

Figure 4.4: Different paraphrasing and preprocessing of sentences (2)

Main sentence = নজরুলের তিন ভাইয়ের মধ্যে কনিষ্ঠ কাজী আলী হোসেন এবং দুই বোনের মধ্যে সবার বড় কাজী সাহেবজান ও কনিষ্ঠ উম্মে কুলসুম।

S1 = নজরুলের তিনটি ভাই ছিলেন: কাজী আলী হোসেন (যিনি সবচেয়ে ছোট), এবং দুইটি বোনের মধ্যে সবচেয়ে বড় ছিলেন কাজী সাহেবজান ও কনিষ্ঠ উম্মে কুলসুম।

S2 = নজরুলের তিনটি ভাই ও দুইটি বোন ছিলেন: কাজী আলী হোসেন (যিনি সবচেয়ে ছোট), এবং সবচেয়ে বড় বোন ছিলেন কাজী সাহেবজান ও কনিষ্ঠ উম্মে কুলসুম।

S3 = নজরুলের তিন ভাই ও দুই বোন ছিল।

S4 = নজরুলের তিনটি ভাই ছিলেন: কাজী আলী হোসেন (যিনি সবচেয়ে ছোট), এবং দুইটি বোনের মধ্যে সবচেয়ে বড় ছিলেন কাজী সাহেবজান ও কনিষ্ঠ উম্মে কুলসুম।

S5 = তিন নাজরুল ভাই হলেন কাজী আলী হোসেন এবং দুই বোনের মধ্যে বড় হলেন কাজী সাহেবজান এবং কনিষ্ঠতম উম্ম কুলসুম।

Figure 4.5: Different paraphrasing and preprocessing of sentences (3)

Main sentence = ১৯০৮ সালে তার পিতার মৃত্যু হয়, তখন তার বয়স মাত্র নয় বছর।

S1 = ১৯০৮ সালে, যখন তার পিতা মারা গেছেন, তখন তার বয়স ছিল মাত্র নয় বছর।

S2 = ১৯০৮ সালে, তার পিতা মারা গেছেন এবং তখন তার বয়স ছিল মাত্র নয় বছর।

S3 = দুর্ভাগ্যবশত, মাত্র নয় বছর বয়সে তার বাবা মারা যান।

S4 = ১৯০৮ সালে, যখন তার পিতা মারা গেছেন, তখন তার বয়স ছিল মাত্র নয় বছর।

S5 = তার বাবা ৭ সালে মারা গিয়েছিলেন, তারপরে তাঁর বয়স ছিল মাত্র নয় বছর।

Figure 4.6: Different paraphrasing and preprocessing of sentences (4)

Main sentence =এ সময় নজরুল মক্তাব থেকে নিম্ন মাধ্যমিক পরীক্ষায় উত্তীর্ণ হয়ে উক্ত মক্তাবেই শিক্ষকতা শুরু করেন।

S1 = এই সময়ে তিনি যেভাবে অত্যাধুনিক মক্তাব থেকে মাধ্যমিক পরীক্ষার উত্তীর্ণ হন এবং একই সময়ে শিক্ষক হিসেবে প্রথম চরণের শিক্ষা শুরু করেন, তার জীবনবৃত্তান্তে একটি শোনানো হয়েছিল।

S2 =এই সময়ে তিনি কিভাবে অত্যাধুনিক মক্তাব থেকে মাধ্যমিক পরীক্ষার উত্তীর্ণ হন এবং একই সময়ে শিক্ষক হিসেবে প্রথম চরণের শিক্ষা শুরু করেন, এই ঘটনাগুলি তার জীবনবৃত্তান্তে একটি অংশ ছিল।

S3 = প্রতিকূলতা সত্ত্বেও, নজরুল তার শিক্ষা অব্যাহত রাখেন এবং এমনকি অল্প বয়সে শিক্ষক হয়েছিলেন।

S4 = এই সময়ে তিনি যেভাবে অত্যাধুনিক মক্তাব থেকে মাধ্যমিক পরীক্ষার উত্তীর্ণ হন এবং একই সময়ে শিক্ষক হিসেবে প্রথম চরণের শিক্ষা শুরু করেন, তার জীবনবৃত্তান্তে একটি শোনানো হয়েছিল।

S5 =এই মুহূর্তে, নাজরুল মক্তাবের নিম্ন মাধ্যমিক পরীক্ষার অনুমোদন দিয়ে মক্তাবীকে পড়াতে শুরু করেছিলেন।

Figure 4.7: Different paraphrasing and preprocessing of sentences (5)

Chapter 5

Impacts of the Project

5.1 Economic Impact

Since our project aims to stop plagiarism through providing a comprehensive platform that caters to the specific needs of various domains. Thus it will have a huge economic impact as authors, journalists, Bangla content creator won't have to think about copyright or plagiarism issues anymore. As a result, their economic background will prosper as much as their morale. Moreover, this positive impact will encourage them to create more original and authentic contents for bengali community, helping to generate more revenue for them and the publishers, thus directly contributing to the national economy too.

5.2 Social and Ethical impact

'SHOTOTA' will also create a positive social impact as it encourages and ensures originality and promotes honesty. "SHOTOTA" will break the unethical social norms in bengali society by creating a new foundation with its root embedded deeply with honesty and originality. Starting from education sector to publishing sector "SHOTOTA" will act as a morale police by stopping plagiarism and promoting an ethical environment among bengali content creators, students and mass people. Thus, its social and ethical impact will be eye catching and tremendous.

Chapter 6

Project Planning

Our CSE499 project spanned from February to November 2023. In the initial phase (Feb-Apr), we extensively researched our project, aiming to identify the main problem. Moving into May and June, we collectively developed a simple version of our project to showcase its functionality.

In July, our focus shifted to documenting our progress in the form of a comprehensive report (CSE499A). Following this, from August to September, we collaborated further to enhance and complete our project. October was earmarked for thorough testing and addressing any issues that arose.

As a team, we approached each month with a set of specific tasks, ensuring a steady and progressive workflow. By November, we aimed to culminate our efforts by crafting the final report summarizing our project's journey, achievements, and outcomes. This structured plan was designed to facilitate steady progress and enabled us to successfully complete our project on time.

Task Name	CSE499A					CSE499B				
	Feb23	Mar23	Apr23	Ma23	Jun23	Jul23	Au23	Se23	Oct23	Nov23
Prepare Plan										
Research Literature										
Problem Finding										
Prepare demo project										
Report for 499A										
Vector Embedding										
Prepare full project										
Testing										
Final report										

Figure 6.1: Project Schedule

Chapter 7

Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP) and Project Relevance

7.1.1 Plagiarism Detection in Bangla

Creating the first plagiarism detection system for Bangla involves addressing language-specific challenges and contributing to the development of tools tailored to the Bengali context.

7.1.2 Semantic Paraphrasing

Developing a system capable of handling paraphrased plagiarism is crucial for enhancing the accuracy of plagiarism detection.

7.1.3 Efficient Vector Database Handling

Integrating Qdrant as the vector database addresses the challenge of efficiently managing vector data, ensuring optimal performance and storage capabilities for the plagiarism detection system.

7.1.4 Accuracy vs. Speed Trade-offs

Prioritizing accuracy while considering speed trade-offs is essential.

7.1.5 Large Dataset

Designing the system with scalability in mind, incorporating vector embeddings and indexing strategies, addresses the challenge of handling large datasets effectively, making the plagiarism detection system robust and adaptable.

Number	Complex Engineering Problems (CEP)	Relevance to Project
1	Plagiarism Detection in Bangla	Creating the first Bang plagiarism detection system
2	Semantic Paraphrasing	Developing a system that can handle paraphrased plagiarism
3	Efficient Vector Database Handling	The project addresses this challenge by integrating Qdrant as the vector database
4	Accuracy vs. Speed Trade-offs	The project addresses this challenge by prioritizing accuracy through the use of vector embeddings and fine-tuning the system for efficient results within resource constraints.
5	Large Dataset	The project addresses this challenge by designing the system with scalability in mind, using vector embeddings and indexing strategies to handle large datasets.

Table 7.1: Complex Engineering Problems (CEP).

7.2 Complex Engineering Problem Activities

7.2.1 System Design with Vector Embeddings

- Allocate human and financial resources to design the plagiarism detection system.
- Obtain the necessary tools and technologies specifically for implementing vector embeddings in the system.

7.2.2 Scalable System Architecture

- Apply creative thinking to develop a system capable of handling a growing collection of Bangla texts.

- Utilize engineering skills to construct a scalable architecture, incorporating smart indexing strategies for efficient processing.

Number	Engineering Activity	Attributes Addressing the Complex Engineering Activity
1	System Design with Vector Embeddings	Allocate human and financial resources for designing the plagiarism detection system. Acquire necessary tools and technologies for vector embeddings.
2	Scalable System Architecture	Creativity is essential to build a system that handles an expanding collection of Bangla texts. Engineering skills are needed to construct a scalable architecture with smart indexing strategies for efficient processing.

Table 7.2: Complex Engineering Problem Activities.

Chapter 8

Conclusions

8.1 Summary

In summary, the lack of a reliable Bengali plagiarism detection tool has negatively impacted the educational and publishing sectors in Bangladesh. To address this issue, our proposed Bengali plagiarism detection tool utilizes preprocessing, the BERT model, a vector database, a plagiarism detection algorithm, and a user-friendly interface. By leveraging advanced techniques such as vector embeddings and web scraping, the tool aims to accurately identify instances of plagiarism in Bengali text. It emphasizes factors such as accuracy, speed, scalability, coverage, privacy, and user-friendliness. Preliminary results demonstrate the feasibility of the approach. Overall, the tool aims to enhance academic integrity and maintain the richness of Bengali language and literature.

8.2 Limitations

8.2.1 Adding Large Corpus

The system faces challenges when incorporating a large corpus due to potential resource constraints, both in terms of storage and processing power.

8.2.2 Budget for Continuous Operation

Maintaining the hardware and software system operational 24/7 requires a significant budget. Continuous operation necessitates robust infrastructure and ongoing support, contributing to the overall cost.

8.2.3 Detecting Complex Paraphrasing

The system encounters difficulties in accurately detecting complex paraphrasing. Identifying nuanced variations in language while maintaining efficiency poses a challenge.

8.2.4 MILVUS Database Compatibility

Initially, MILVUS was preferred for its speed and efficiency, but its inability to handle Bangla texts led to a transition to QDRANT, which, although functional, is notably slower and less efficient than MILVUS..

8.3 Future Improvement

8.3.1 Corpus Expansion

Our future strategy includes the expansion of the corpus, allowing us to incorporate a broader range of data for improved analysis and performance.

8.3.2 Indexing Parameter Optimization

We plan to optimize indexing parameters to enhance the efficiency and effectiveness of our system in handling and retrieving information.

8.3.3 Advanced Paraphrasing Detection Algorithm

Implementing a more efficient algorithm to detect complex paraphrasing, enhancing the system's ability to recognize nuanced language variations.

8.3.4 Migration to MILVUS Database

A shift back to the MILVUS database is anticipated upon the release of a new patch capable of handling Bangla texts, aiming for the previously experienced efficiency while accommodating diverse languages.

APPENDIX

Here is the source code: https://bitbucket.org/rhb17/bangla_plagiarism/src/master/

Results of different types of paraphrasing : <https://drive.google.com/drive/folders/1wwJ0UMzCE-d9S5wEn9>

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